

OPERATING SYSTEMS — EXAM REVISION GUIDE

Prioritized Slide Ranking & Core Exam Topics (Ranked from Most Important to Lowest)

TIER 1: HIGHEST PRIORITY (CRITICAL EXAM TOPICS — ~60-70% WEIGHTAGE)

CH-05.pdf — Process Scheduling [Rank 1 (Most Important)]

- **CPU Scheduling Algorithms (Gantt Charts & Numericals):** First-Come First-Served (FCFS), Shortest Job First (SJF), Shortest Remaining Time First (SRTF / Preemptive SJF), Preemptive Priority Scheduling, Round Robin (RR) with time quantum.
- **Performance Metrics:** Calculations of Average Waiting Time (AWT), Average Turnaround Time (ATT), Response Time, Throughput, and CPU Utilization.
- **Concepts & Issues:** Preemptive vs. Non-preemptive scheduling, Starvation problem, Convoy Effect, and solutions like Aging.

CH-08 1.pdf & CH-08 2.pdf — Deadlocks [Rank 2]

- **Banker's Algorithm (Numericals):** Calculating Need Matrix ($\text{Need} = \text{Max} - \text{Allocation}$), Safety Algorithm to find Safe State / Safe Sequence, and Resource-Request Algorithm to grant/deny process requests.
- **Deadlock Characterization:** 4 Coffman Conditions (Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait).
- **Resource Allocation Graph (RAG):** Constructing RAGs and detecting deadlocks/cycles.
- **Deadlock Strategies:** Prevention, Avoidance, Detection, and Recovery methods (Process Termination vs. Resource Preemption).

CH-12.pdf — Mass-Storage Systems & Disk Scheduling [Rank 3]

- **Disk Scheduling Algorithms (Numericals):** FCFS, SSTF, SCAN (Elevator), C-SCAN, LOOK, and C-LOOK. Calculating total head movement / seek distance in cylinders.
- **Disk Performance Parameters:** Seek Time, Rotational Latency, Transfer Time, Disk Access Time, and Disk Response Time.
- **Concepts:** Purpose of disk scheduling, criteria for selecting scheduling algorithms, and physical disk structure (platters, tracks, sectors, cylinders).

CH-10.pdf — Virtual Memory [Rank 4]

- **Page Replacement Algorithms (Numericals):** First-In-First-Out (FIFO), Optimal Algorithm, Least Recently Used (LRU). Counting page faults for a reference string given fixed frame allocations.
- **Segmented Paging Address Translation:** Translating Logical Address (Segment Number, Offset) to Physical Address using Segment Table and Page Table.
- **Core Concepts:** Page Fault handling steps, Valid-Invalid Bit, Thrashing (causes, effects on CPU utilization, prevention/recovery), Belady's Anomaly.

TIER 2: HIGH PRIORITY (CORE THEORETICAL & ALGORITHMIC TOPICS)

CH-06 1.pdf & CH-06 2.pdf — Synchronization Tools [Rank 5]

- **Critical Section Problem:** Evaluating lock mechanisms against Mutual Exclusion, Progress, and Bounded Waiting.
- **Peterson's Solution:** Software synchronization for two processes using turn and interested variables.
- **Semaphores:** Binary Semaphore (Mutex) vs. Counting Semaphore; Wait (P) and Signal (V) operations.
- **Classic Problems:** Producer-Consumer Problem and Reader-Writer Problem solutions using semaphores.

CH-09.pdf / CH-09_2.pdf — Main Memory Management [Rank 6]

- **Dynamic Storage Allocation Algorithms:** First-fit, Best-fit, and Worst-fit partition allocation.
- **Memory Fragmentation & Management:** Internal vs. External Fragmentation, Compaction / Defragmentation technique, Fixed vs. Dynamic Partitioning.
- **Address Binding & Hardware Protection:** Compile time, Load time, Execution time binding; Memory Management Unit (MMU), Base & Limit / Relocation registers.

TIER 3: MEDIUM PRIORITY (CONCEPTUAL THEORY & SHORT QUESTIONS)

CH-03.pdf — Processes [Rank 7]

- **Process Control Block (PCB):** Structure and stored information (Process State, PC, Registers, CPU Scheduling info, Memory info).
- **Process Management:** Process States (New, Ready, Running, Waiting, Terminated) and State Transition Diagrams.
- **Context Switch & Schedulers:** Context switching mechanism, overheads, and comparison between Short-term, Long-term, and Medium-term schedulers.
- **Inter-Process Communication (IPC):** Shared Memory Model vs. Message Passing Model.

CH-04.pdf — Threads [Rank 8]

- **User vs. Kernel Threads:** Comparison, advantages, disadvantages, and implementation models.
- **Multithreading Concepts:** Benefits of multithreading (Responsiveness, Resource Sharing, Economy, Scalability).
- **Thread Architecture:** Shared process components (Code, Data, OS Resources) vs. Private thread components (Program Counter, Registers, Stack).

TIER 4: LOWER PRIORITY (INTRODUCTORY THEORY & DEFINITIONS)

CH-01.pdf — Introduction to Operating Systems [Rank 9]

- **Dual-Mode Operation:** User Mode vs. Kernel Mode, Mode Bit, Mode transitions via System Calls.
- **Interrupts & Traps:** Hardware Interrupt vs. Software Trap/Exception, Interrupt Vector, Interrupt Service Routine (ISR), Interrupt-driven I/O cycle.
- **System Architecture:** Direct Memory Access (DMA), Storage Hierarchy (Registers, Cache, Main Memory, Secondary Storage), Open-Source OS concepts.

CH-02.pdf — Operating-System Structures [Rank 10 (Lowest Relative Priority)]

- **System Calls:** Definition, System Call Interface, System Call sequence for file operations (e.g., copying/moving files).
- **OS Architectures:** Monolithic, Microkernel (benefits/detriments), Layered, Modular, and Hybrid structures.
- **System Booting:** Bootstrap Program role and initialization sequence.